BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 243 · Year 13 autumn mock: Paper 1

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. Evaluate true AND false.

Q2. Why should a model omit irrelevant details?

Q3. What does the Program Counter hold?

Q4. A full adder receives A=1, B=1 and carry-in=1. State sum and carry-out and justify using binary addition.

Do Now Recall previous learning | **I Do** Read new knowledge | **We Do** Practise with guidance | **You Do** Work independently | **Review & Improve** Check understanding

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I Do · Read new knowledge

Key vocabulary

diagnostic — Assessment used to find knowledge gaps.

two's complement — Encoding signed binary integers.

De Morgan's law — Negate operands and swap AND/OR.

register — Fast CPU storage for a specific role.

New learning

Purpose — This is an original diagnostic Paper 1 practice, not an official OCR examination paper. Some systems topics will be retaught later.

Manage time — Allow approximately 15 minutes to each 15-mark question.

Show methods — Write the inversion/add-one steps, state every Boolean simplification rule and use precise CPU register terminology.

Worked example

Unmarked example: what earns method marks for a negative 8-bit value?

1. Predict the method — Write +5 in exactly 8 bits.

2. Check and explain — Invert each bit, then add one to obtain the two's complement result for −5.

We Do · Practise with guidance

Unmarked different example: evaluate ¬(A ∧ B) for A=1, B=0.

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A ∧ B = 1 AND 0 = 0
¬0 = 1
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Predict the method

First calculate A AND B = 0.

Check and explain

Negate it: ¬0=1. Include intermediate reasoning when a question asks for it.

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2. Write a complete new answer in GREEN PEN. Show working if needed:

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You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

(a) Convert −18 to eight-bit two's complement, showing working. [4] (b) Perform arithmetic right shift by 1 on that representation, giving the binary result and signed decimal value. [3] (c) Add 1011_2 and 0110_2, showing carry and answer. [3] (d) Normalise positive $0.0011_2 \times 2^5$, showing shifts and decimal value. [5]

Question 2

(a) Complete four output rows of A XOR B in order 00,01,10,11. [4] (b) Simplify ¬(A ∧ B) using De Morgan's law. [3] (c) For a half adder with A=1, B=1, state Sum and Carry and explain the gates. [4] (d) Simplify A ∨ (B ∧ ¬B) using Boolean identities. [4]

Question 3

(a) State the roles of PC, MAR, MDR, CIR and ACC. [5] (b) Describe a simplified fetch-decode-execute cycle using PC, MAR, MDR, CIR and the control unit. [6] (c) Explain one role of an interrupt and an interrupt service routine. [4]

Exit Ticket · Answer before checking

For an 8-bit signed representation, adding a ninth bit automatically gives the correct 8-bit answer.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

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Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. False.

Q2. To reduce complexity while retaining data needed to answer the question.

Q3. The address of the next instruction to fetch.

Q4. 1+1+1=3 decimal=11 binary; Sum=1, carry-out=1.

You Do pages · Independent questions

Q1. MARK SCHEME /15. (a) +18=00010010 [1], invert 11101101 [1], add 1=11101110 [2]. (b) Sign-preserving right shift gives 11110111 [2] = −9 [1]. (c) Correct binary addition yields 10001_2 [2], correct carry evidence [1]. (d) $0.0011\times2^5 \rightarrow 0.0110\times2^4$ [2] $\rightarrow 0.1100\times2^3$ [2], decimal 6 [1].

Q2. MARK SCHEME /15. (a) Outputs 0,1,1,0 [4]. (b) ¬A ∨ ¬B [2], De Morgan swaps AND to OR and negates components [1]. (c) Sum=0 [1], Carry=1 [1], XOR gives sum [1], AND gives carry [1]. (d) B∧¬B=0 [2]; A∨0=A [2].

Q3. MARK SCHEME /15. (a) PC address of next instruction [1], MAR memory address accessed [1], MDR data/instruction transferred [1], CIR current instruction [1], ACC result/intermediate arithmetic [1]. (b) PC address into MAR [1]; instruction fetched from memory into MDR [1]; PC advanced [1]; instruction copied to CIR [1]; control unit decodes [1]; processor executes and updates registers as needed [1]. (c) Interrupt signals event requiring attention [1]; CPU checks/services it at a suitable point [1]; relevant ISR handles event [1]; execution state restored and normal processing resumes [1].

Exit Ticket · Model answer

False. Preserve the specified width and account for overflow.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question: